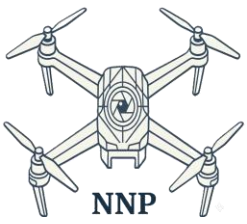


SMART INDIA HACKATHON 2026



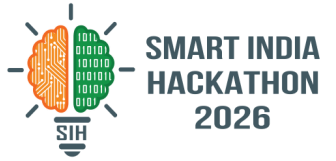
- **PROBLEM STATEMENT ID – SIH26177**
- **PROBLEM STATEMENT TITLE-** A deployable AI-powered autonomous drone that aids search-and-rescue operations by detecting people and hazards, thereby improving responder safety and reducing victim discovery time.
- **THEME – DISASTER MANAGEMENT**
- **PS CATEGORY-** Hardware
- **TEAM ID - 152087**
- **Team Name - NNP**





NNP

On-device air-ground swarm for autonomous survivor search and rescue routing



Problem at Hand

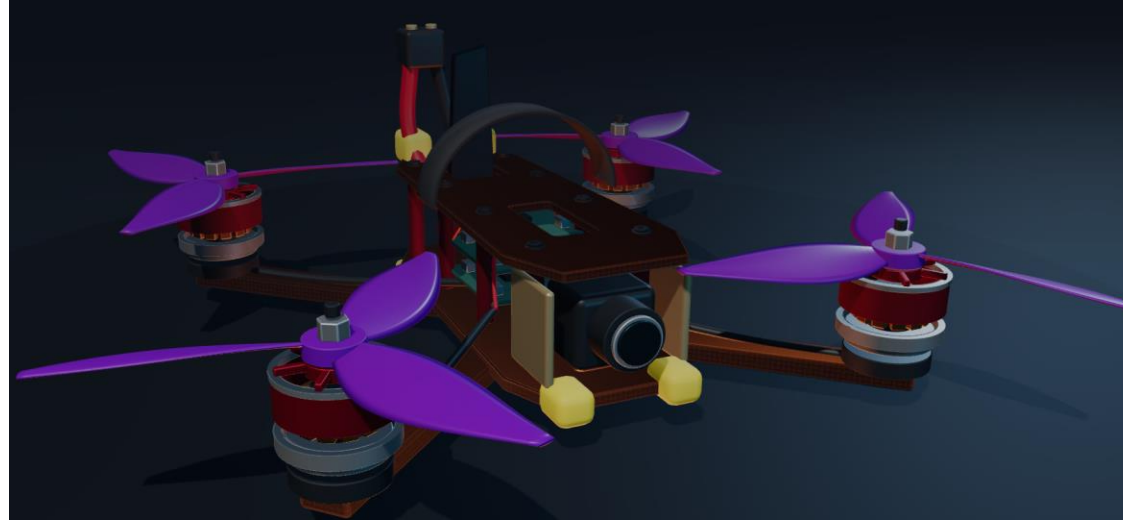
Immediate Access: Disasters like structural collapse or flood cut off immediate rescue access, making first hours critical.

Time Criticality: Survival rates drop sharply after 72 hours; every minute of search time saves lives.

Sensor Limitations: Standalone visual or thermal cameras on single drones struggle to identify life through dense debris or multiple layers.

Communication Gaps: Standard networks fail in disaster zones; single drones cannot relay heavy sensor data back to base over long distances.

Our Solution



Why We Stand Out

- **Resilient Communications:** Self-healing mesh ensures robust data relay from deep disaster sites, unlike single-drone setups.
- **OUR OWN GPS:** no GPS no problem, Our own network gives hexapods and drones a local positioning system that works without satellite GPS, so hazard boundaries are set and enforced accurately even inside collapsed structures and signal-blocked zones.
- **Ground-Truth Ground Probing:** Ground hexapods adaptively assess terrain for true human traversability, unlike purely aerial

Key Features



Multi-Modal AI Detection (AI Cam): Combined YOLO and SAM analysis on RGB and Thermal feeds to map threats and detect anomalies.



Thermal Deep Scan (Heat Sensor): Radiometric analysis of small heat pockets, distinguishing human body heat (37° C) from surrounding fires.



mmWave Bioradar (mmWave Bioradar): Life sign verification by detecting microscopic 0.2-0.6 Hz respiratory rhythms through rubble.



Acoustic SOS Detection (Acoustic SOS Detection): Noise-cancelled audio array to isolate human distress calls and tapping over drone motor noise.



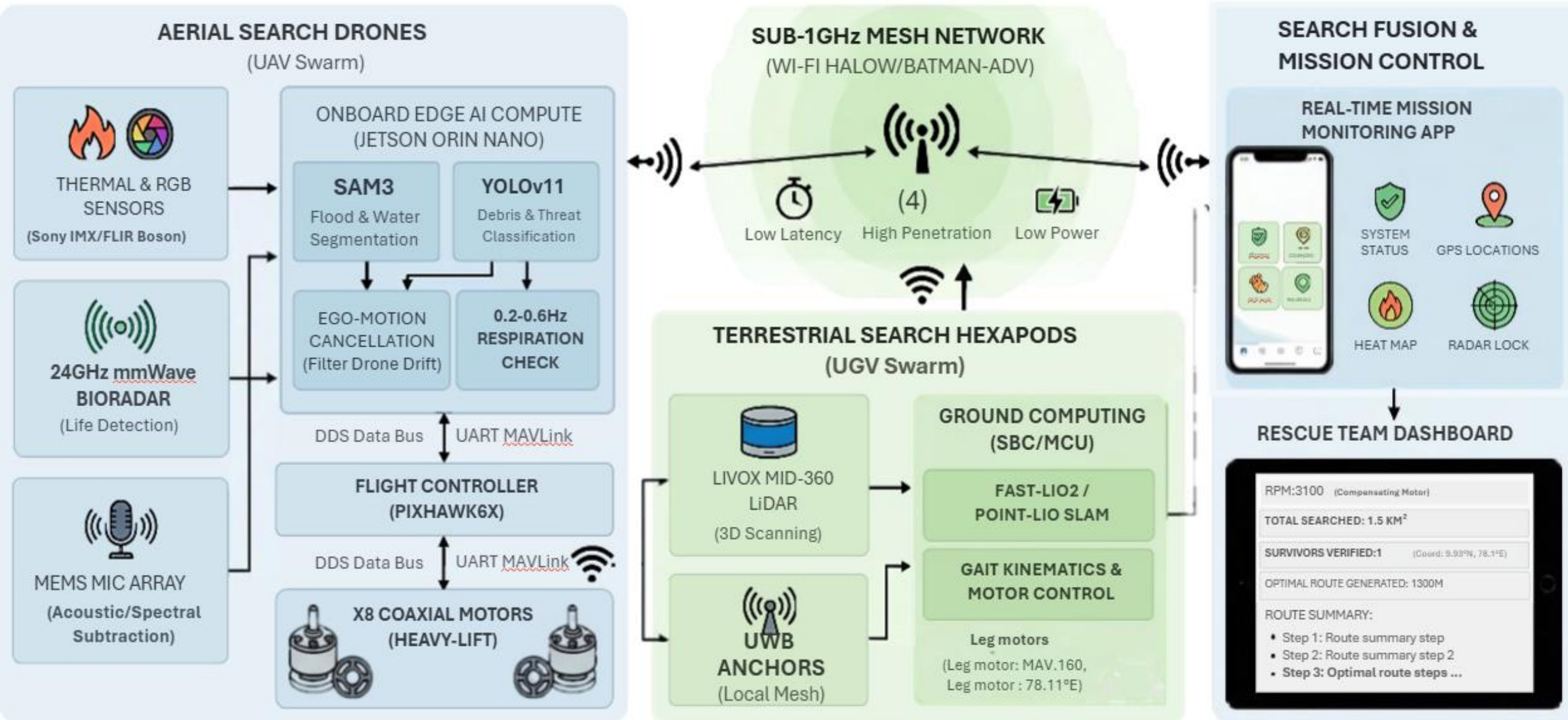
Legged Terrain Probe (Legapod Leg Probe): Ground hexapods probe surface stability and clearance, assessing human traversability.



Self-Healing Mesh Relay (Self-Healing Mesh): Transparent BATMAN-adv mesh over 900 MHz provides deep penetration and uninterrupted data from multiple views.

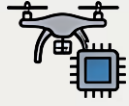


TECHNICAL APPROACH





FEASIBILITY AND VIABILITY



Technical Feasibility

- Built using **high-performance COTS** components: NVIDIA Jetson Orin Nano (Edge AI), Holybro Pixhawk 6X (Flight Control), and robust coaxial X8 frames.
- Integrates **multi-modal sensors** (Sony RGB, FLIR Thermal, 24GHz Bioradar, MEMS Mic, Livox LiDAR) with advanced AI models (YOLO-v11, SAM 3, FAST-LIO2) for robust detection.
- Utilizes **self-healing B.A.T.M.A.N.** Advanced Sub-1GHz mesh network for resilient low-latency communication in infrastructure-denied zones.



Economic and Operational Viability

- High initial investment is justified by immense value in lives saved and drastically reduced search times versus traditional methods.
- Total pilot swarm cost estimated at ₹15-20 Lakhs, but per-mission operational cost is low (battery charging, minimal field maintenance).
- Modular, scalable design allows for expanding swarm size and adapting sensor payloads for diverse disaster scenarios.

Economical Feasibility

Evaluation Parameter	Current Manual / Traditional SAR	Proposed Smart System (AeroGrid)
Cost of Operation	Variable, High (Manpower & Logistics)	Fixed Initial, Low Operational
Manpower Required	Large Teams (dozens to hundreds)	Small Specialist Team (3-5 operators)
Daily Coverage Area	Slow, limited by terrain & safety	Rapid, Wide-Area Swarm Coverage ~1.5km ² /hr
Detection Accuracy	Lower, weather/terrain dependent	High (Multi-sensor fusion & AI)
Response Time	Slow (hours to days for mobilization)	Rapid (Immediate deployment)
Power Consumption	High (Fuel for vehicles/aircraft)	Moderate (Rechargeable Batteries)

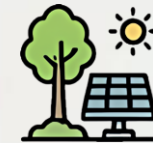
Implementation Viability

Factor	Support Element
Manufacturability	Standard drone/robotics integration; frames & parts are COTS or easily fabricated.
Component Availability	Most critical components (Jetson, Pixhawk, Sensors) are globally sourced COTS with high availability.
Serviceability	Modular design allows for quick field-swapping of arms, motors, propellers, and sensor payloads.
Scalability	Swarm architecture is inherently scalable; simply add more nodes to increase coverage.
Deployment Readiness	Prototype & sub-system validation complete; integrated system ready for controlled field trials within 6-9 months.



Maintenance & Safety Feasibility

- Field-serviceable design: modular components (motors, props, batteries) allow for quick repairs between missions.
- Multiple redundancy levels** (dual GPS, triple IMU on Pixhawk) and failsafe protocols (RTL, emergency landing) ensure safe operation.
- Operational safety: APF for collision avoidance, dynamic geofencing, and integrated health monitoring provides pre-emptive failure alerts.

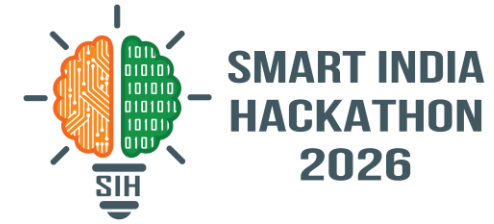


Environmental Viability

- Zero-emission operation during flight/ground movement due to all-electric propulsion systems.
- Minimal physical environmental impact** Drone fly above terrain, and hexapods have low ground pressure.
- Aligns with disaster resilience goals and promotes indigenous technology development (**Atmanirbhar Bharat**) for critical national capabilities.



IMPACT AND BENEFITS



Minimal Discovery time



victims fully buried or hidden from camera are found using radar and thermal view

Immediate responder casualties



Survivors are found and rescued in significantly less time

Operations in zero-infrastructure



search and detection continue fully offline, exactly where and when most SAR technology fails

Eliminates single points of failure



Mission keeps running even if a drone or communication link is lost mid-operation

Multi-layered detection redundancy



RGB, thermal, acoustic, RF, and radar together mean a missed signal in one channel is caught by another — no single sensor failure costs a life

Public trust in disaster response



Making it a technology that communities and authorities can genuinely rely on and accept

Enforcement of hazard boundaries



A six-legged hexapod unit navigates unstable rubble terrain to physically enforce geofenced hazard boundaries in real time

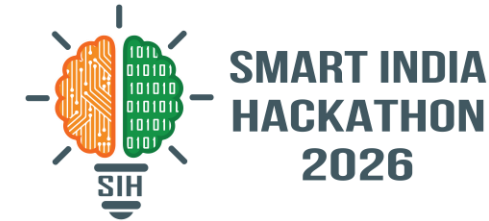
Lower cost per life saved



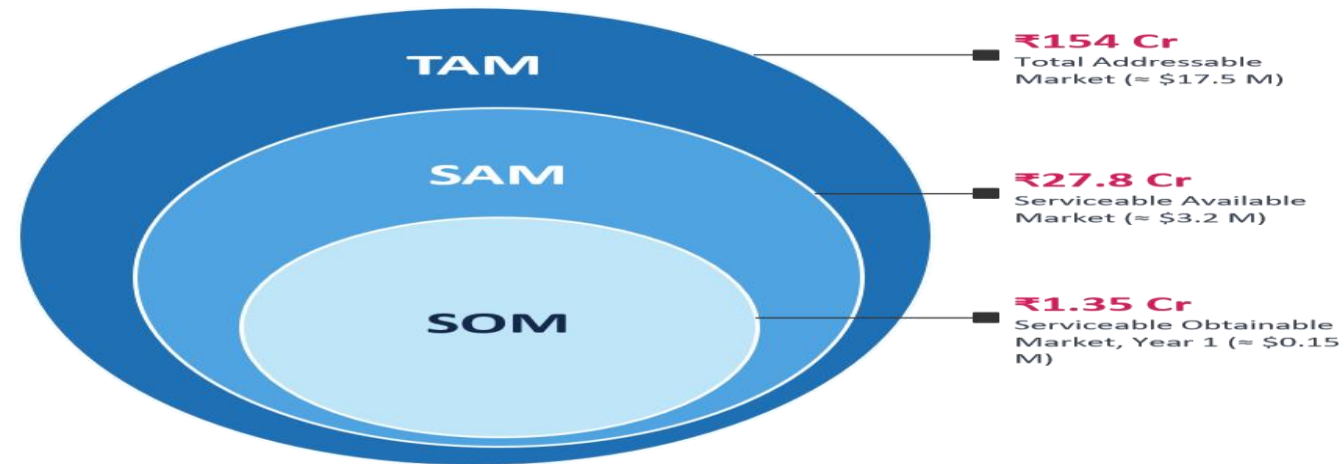
A swarm of low-cost drones and ground units covers more ground with less manpower and equipment



RESEARCH AND REFERENCES



Market Opportunity: TAM, SAM, SOM



- Pramudita, A. A. et al., "Radar System for Detecting Respiration Vital Sign of Live Victim Behind the Wall," IEEE Sensors Journal, 2022 (higher-tier journal version) <https://ieeexplore.ieee.org/document/9826342/>
- Accurate Sensing of Multiple Humans Buried Under Rubble Using IR-UWB SISO Radar During Search and Rescue," Sensors and Actuators A: Physical, 2022 <https://www.sciencedirect.com/science/article/abs/pii/S0924424722006100>
- Review — Microwave Radar Sensing Systems for Search and Rescue Purposes," PMC, NIH <https://www.ncbi.nlm.nih.gov/pmc/articles/PMC6650952/>
- A Bio-Inspired Swarm UAV Framework Integrating Thermal Sensing and Optimization-Based Coordination for Efficient Search and Rescue Operations," Scientific Reports (Nature), 2025 <https://www.nature.com/articles/s41598-025-33223-z>
- Bartolomei, L., Teixeira, L., Chli, M., "Towards Multi-Robot Exploration: A Decentralized Strategy for UAV Forest Exploration," ETH Zürich (arXiv) <https://arxiv.org/pdf/2301.08537>
- Coordinated Search With a Swarm of UAVs
- https://www.academia.edu/303273/Coordinated_Search_With_a_Swarm_of_UAVs

Company/industry references :

- ideaForge x C-DAC — Drone-as-a-Service integration with India's Emergency Response Support System (ERSS) <https://dronelife.com/2025/12/11/ideaforge-and-c-dac-partner-to-integrate-drone-as-a-service-platform-with-indias-emergency-response-system/>
- Garuda Aerospace — deployed drones with NDRF for the SLBC tunnel collapse rescue, Telangana, 2025 <https://dronelife.com/2025/02/25/garuda-aerospace-drones-aid-ndrf-in-telangana-tunnel-rescue/>

➤ [github repository](#)

➤ [youtube video](#)

Destination	Direct Production URL
3D Hexapod Model Viewer	https://aegisdisasterworld.vercel.app/hexapod-viewer/ (or /hexapod)
3D Drone Model Viewer	https://aegisdisasterworld.vercel.app/drone-viewer/ (or /drone)
3D Incident Command Simulation	https://aegisdisasterworld.vercel.app/
NNP Mission Control Dashboard	https://aegisdisasterworld.vercel.app/nnp-dashboard/